

AMMAR

Machine Learning / AI

Q3 — Natural Language Processing: Word Cloud Analysis

Objective

To apply Natural Language Processing (NLP) on a Twitter dataset to classify tweets as Positive or Negative, then visualize the most frequent words in each category using Word Clouds — implemented in Google Colab.

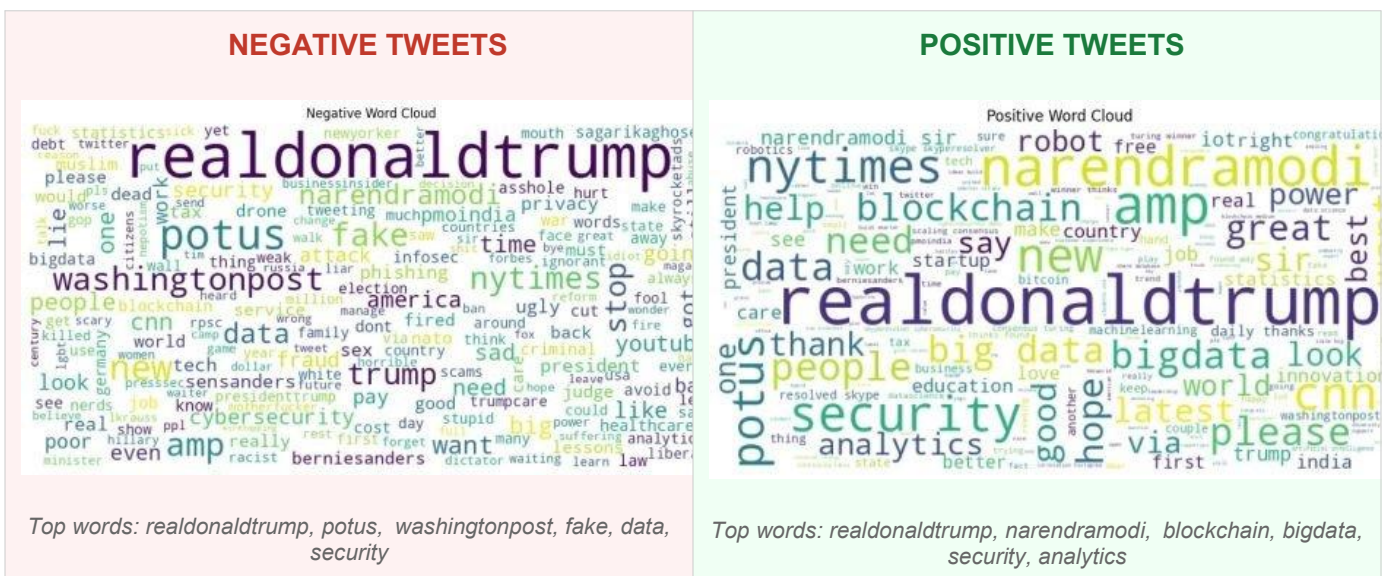
Tools & Libraries

Platform	Google Colab (cloud-based Python notebook)
Libraries	NLTK, TextBlob / VADER, WordCloud, Matplotlib, Pandas
Dataset	Twitter dataset with labeled Positive and Negative tweets

Process Steps

- Text Preprocessing — removed URLs, @mentions, stopwords, and special characters
- Sentiment Labeling — tweets classified as Positive (1) or Negative (0)
- Word Frequency — most common words extracted per sentiment class
- Visualization — Word Clouds generated with custom color maps using WordCloud library

Results — Word Cloud Visualization



Conclusion: NLP sentiment analysis successfully separated positive and negative tweet patterns. The Word Cloud visualization confirmed clear thematic differences between the two sentiment classes.